

TIC 150230753

Eclipsing binary candidate dossier - Star Vetter

PERIOD ALIAS

BLS EPHEMERIS		CATALOG CROSS-MATCH (novelty)	
Period	2.334152 d (ALIASED, see below)	VSX	no match
-> corrected period:	4.668305 d	ASAS-SN	no match
Epoch (t0)	1626.0382 BTJD	Gaia eclipsing-binary table	no match
Duration	0.1200 d	Gaia variable classification	no match
Depth	16.455 % (164,549 ppm)	TESS EB catalog (Prsa+2022)	no match
SNR	235.7	ExoFOP TOI list	no match
INDEPENDENT CHECKS (f3i gates)		Verdict: NOVEL	
Eclipses observed	7 (gate >= 3)		
Robust depth ratio	1.01 (gate >= 0.50)		
Robust depth sigma	53.0 (gate >= 7.0)		
Odd/even z-score	4.17 (ALIAS FLAGGED)		
Secondary eclipse	not detected		

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

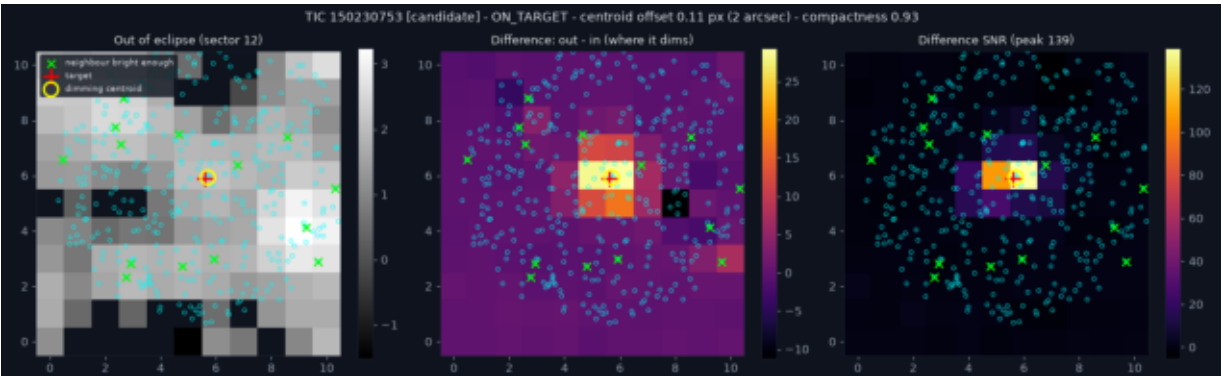
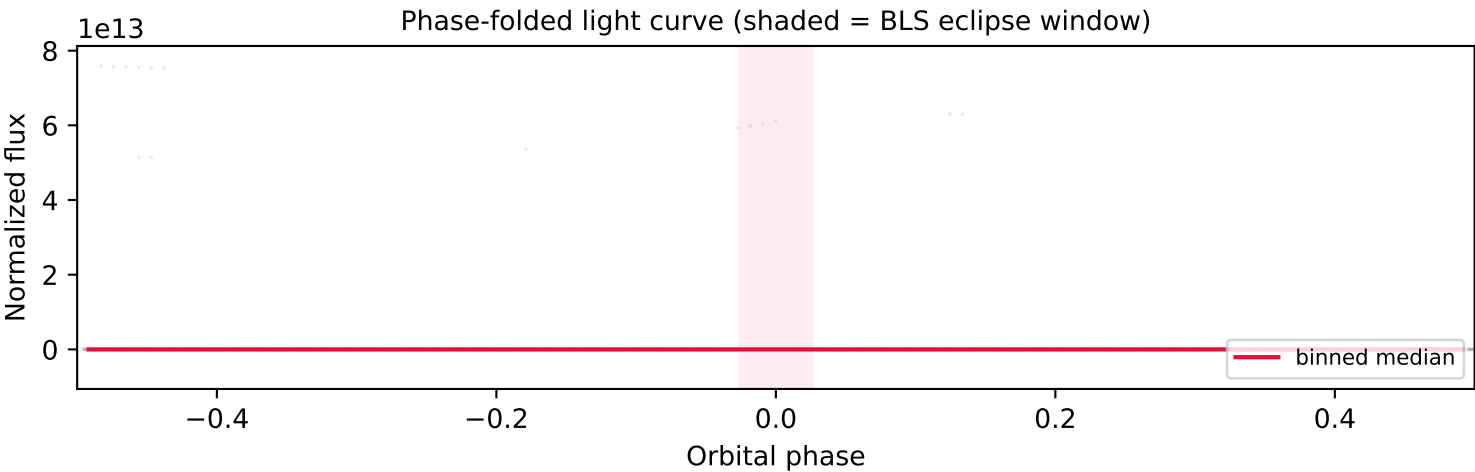
Sector 12 light-curve dip in this sector: 50.2 sigma t0 refined by -0.0075 phase

Difference-image peak SNR 139.03 (gate >= 5.0) Compactness 0.93 (gate >= 0.50)

Dimming centroid offset 0.11 px (2 arcsec) (gate <= 1.0 px, i.e. on the target)

Nearest bright-enough neighbour in cutout: TIC 150237925

1.16 px away, dTmag +0.72 - farther from the dimming than the target



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 4.17 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 4.668305 d, in any write-up.